bias_expert.py

`src/moe_pruning/pruning/bias_expert.py · 46 lines · module moe_pruning.pruning.bias_expert`

What it does

`BiasExpert` replaces a removed expert with one trainable vector b. For every routed token it returns b; the parent MoE layer then multiplies it by the router gate exactly once, giving g * b.

How it works

- The vector is an `nn.Parameter`, so it can be trained; the output is a broadcast copy in the input's dtype.
- For half-precision models the parameter is kept in float32 so that small optimiser steps are not rounded away.

Notes

- The class knows nothing about any architecture; it only needs the output width.

Defined in this file

Extracted from the source code (classes, public functions and methods).

Name	Description
`class BiasExpert`	Trainable constant replacement b for a removed expert (gate applied by the parent).
`.forward(hidden_states)`	Return b broadcast to `hidden_states.shape[:-1] + (output_dim,)` (un-gated).

Connections

Used by: `scripts/train_biases.py`, `src/moe_pruning/pruning/replacement.py`, `src/moe_pruning/training/bias_trainer.py`